

Adaptive RF Modulation Classification Using Residual Denoising and Raw-Denoised Signal Fusion with Residual CNN-Transformer-Attention Learning

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Abstract. Automatic Modulation Classification (AMC), a fundamental task in RF signal recognition, is a critical component in intelligent wireless communication architectures, cognitive radio networks, spectrum monitoring, and adaptive RF applications [1], [2]. However, the predictive capability of conventional deep learning-based AMC pipelines degrades significantly under interference-heavy scenarios where modulation-specific temporal patterns become highly distorted [3], [4]. To address this challenge, this study introduces an Adaptive RF Signal Recognition pipeline using Residual Noise Suppression and Raw-Denoised Signal Fusion with a Residual CNN-Transformer encoder-Attention architecture. The developed pipeline first employs a Residual Noise Suppression Autoencoder (RDAE) to suppress channel noise while preserving significant modulation-specific structures. Instead of replacing the original signal, the denoised RF representation is first combined with the original RF signal using weighted fusion, and the resulting representation is concatenated with the original RF signal to construct a four-channel tensor. The fused representation is then processed utilizing residual CNN representation learning and Transformer encoder-based self-attention learning to capture long-range temporal dependencies across RF signal sequences. Evaluation findings on the RadioML 2016.10a benchmark dataset [10] under practical wireless conditions with $\text{SNR} \geq -8$ dB illustrate that the developed pipeline attains an overall recognition accuracy of 87.20%, surpassing baseline CNN and CNN-BiLSTM architectures. The findings indicate that the fusion-driven residual learning pipeline enhances noise-resilience and RF signal recognition capability under interference-heavy scenarios. Furthermore, the proposed architecture maintains a competitive computational profile with an average inference time of 5.75 ms per frame. These results, coupled with the pipeline's noise resilience, validate its feasibility for real-time deployment in practical cognitive radio and automated spectrum monitoring systems.

Keywords: Automatic RF Signal Recognition, Residual Noise Suppression, RF Signal Processing, CNN, Transformer encoder Attention, Signal Fusion, deep learning, Wireless Communication, RadioML 2016.10a.

1 Introduction

Automatic Modulation Classification (AMC), a fundamental task in RF signal recognition, is an essential technology in modern wireless communication architectures and intelligent radio applications [11], [12]. AMC acts as an intermediate stage between signal detection and signal demodulation, enabling communication architectures to automatically identify modulation formats without prior transmitter knowledge. This capability is highly significant in applications such as cognitive radio, adaptive spectrum allocation, military surveillance architectures, electronic warfare, software-defined radio architectures, and intelligent spectrum monitoring pipelines [13], [14].

Conventional AMC architectures mainly rely on two categories of strategies: Likelihood-Based (LB) approaches and Feature-Based (FB) approaches [15]. Likelihood-based approaches calculate probability distributions for different modulation hypotheses and select the most probable modulation scheme. Although these approaches are theoretically optimal, they require accurate channel estimation and involve extremely high computational complexity. Feature-based approaches rely on handcrafted signal characteristics such as higher-order statistics, spectral features, cyclostationary properties, and amplitude variations [16]. However, handcrafted features fail under interference-heavy scenarios and non-linear wireless environments where modulation patterns become highly distorted.

Recent advances in deep learning have transformed RF signal recognition by enabling automatic representation learning directly from raw I/Q RF samples [1]. CNN-based architectures efficiently learn local spatial and temporal correlations, while sequential architectures such as LSTMs and GRUs capture long-term temporal dependencies across RF signal sequences [12]. Hybrid CNN-LSTM architectures have shown promising predictive capability for recognition tasks. Nevertheless, conventional deep learning networks still experience significant predictive capability degradation under severe noise conditions because channel noise obscures significant modulation-specific structures and temporal transitions [3], [10].

Several noise suppression-driven strategies have been introduced to enhance RF recognition noise-resilience under interference-heavy scenarios [9]. Conventional noise suppression autoencoders reconstruct clean signals from noisy RF samples utilizing encoder-decoder architectures. However, these strategies often introduce over-smoothing effects that remove significant phase transitions and subtle amplitude variations required for distinguishing higher-order modulation schemes such as QAM16 and QAM64 [3].

To address these limitations, throughout this research, a developed Adaptive RF Signal Recognition pipeline is introduced that integrates Residual Noise Suppression, Raw-Denoised Signal Fusion, Residual CNN representation learning, and Transformer encoder-based attention-driven learning. The developed pipeline preserves significant modulation-specific structures while efficiently suppressing channel noise, resulting in enhanced recognition predictive capability under practical wireless environments.

The major contributions of this study are summarized as follows:

1. A Residual Noise Suppression Autoencoder (RDAE) is introduced for RF restoration while preserving modulation-specific temporal structures.
2. A Raw-Denoised Signal Fusion strategy is introduced by applying weighted fusion between raw and denoised RF representations, followed by concatenation with the original RF signal to construct a four-channel tensor structure.
3. A Residual CNN-Transformer encoder-Attention recognition backend is developed for noise-resilient temporal representation learning and long-range dependency learning.
4. Evaluation findings demonstrate enhanced recognition predictive capability compared to baseline CNN and CNN-BiLSTM architectures.

2 Related Work

2.1 Deep Learning for RF Signal Recognition

Deep learning has emerged as one of the most significant research directions for RF signal recognition. Earlier research for RF recognition architectures relied heavily on handcrafted signal features extracted manually from received RF signals. However, these strategies lacked noise-resilience and generalization capability under practical wireless conditions [5]. The introduction of Convolutional Radio Recognition Networks by Tim O'Shea illustrated that CNN architectures are able to automatically learn modulation-specific features directly from raw I/Q RF samples [1], [10]. Following this work, several researchers explored hybrid CNN-LSTM and CNN-GRU architectures [12]. More recently, Transformer encoder-based architectures have gained significant attention because of their ability to model long-range dependencies utilizing self-attention mechanisms [1], [2].

2.2 Residual Learning and RF restoration

Residual learning strategies allow deep learning networks to learn correction mappings instead of complete transformations, resulting in enhanced gradient flow and stable optimization. Noise suppression Autoencoders (DAEs) have been extensively adopted for RF restoration [9]; however, direct signal reconstruction often removes high-frequency signal structures. The developed pipeline adapts residual learning principles to RF restoration by predicting only the noise residual component, preserving significant signal characteristics while efficiently reducing additive channel noise [1], [10].

2.3 Attention-driven learning in RF Signal Processing

Attention-driven learning has revolutionized sequence modeling tasks. The Transformer encoder architecture introduced multi-head self-attention mechanisms able to learn contextual dependencies across long sequences [1]. In RF signal recognition, attention-driven learning allows networks to focus on significant

symbol transition regions and modulation-specific temporal features [2]. Beyond general surveys, recent implementations have demonstrated the efficacy of specialized Transformer variants in the RF domain. For instance, multi-scale CNN-Transformer architectures (such as MCTNet) have been proposed to capture multi-resolution temporal features [17, 19], while others have successfully adapted Vision Transformers (ViT) and attention-based blocks for classification via constellation representations [18, 20]. Our work distinguishes itself by integrating a raw-denoised signal fusion stage before the attention mechanism, ensuring the Transformer focuses on reconstructed structural patterns rather than high-frequency noise residuals.

3 Implementation Strategy

3.1 Signal Representation

The received RF signal is modeled as:

$$r(t) = s(t) * h(t) + n(t) \quad (1)$$

where $s(t)$ represents the transmitted signal, $h(t)$ represents the wireless channel response, and $n(t)$ represents Additive White Gaussian Noise (AWGN). The channel noise is often characterized by the Signal-to-Noise Ratio (SNR), defined mathematically in the decibel scale as:

$$SNR_{dB} = 10 \log_{10} \left(\frac{\sum_{n=1}^L |s[n]|^2}{\sum_{n=1}^L |n[n]|^2} \right) \quad (2)$$

The RF signal is represented utilizing In-phase and Quadrature (I/Q) RF samples: $X_{raw} \in \mathbb{R}^{2 \times 128}$.

3.2 Residual Noise Suppression Autoencoder (RDAE)

The introduced Residual Noise Suppression Autoencoder learns residual noise correction instead of reconstructing the entire signal directly. The denoised signal is obtained as:

$$X_{denoised} = X_{raw} + R(X_{raw}) \quad (3)$$

where $R(X_{raw})$ represents the predicted residual noise component. The optimization of the RDAE is governed by the Mean Squared Error (MSE) loss function:

$$\mathcal{L}_{RDAE} = \frac{1}{N} \sum_{i=1}^N \|X_{clean}^{(i)} - (X_{raw}^{(i)} + R(X_{raw}^{(i)}))\|^2 \quad (4)$$

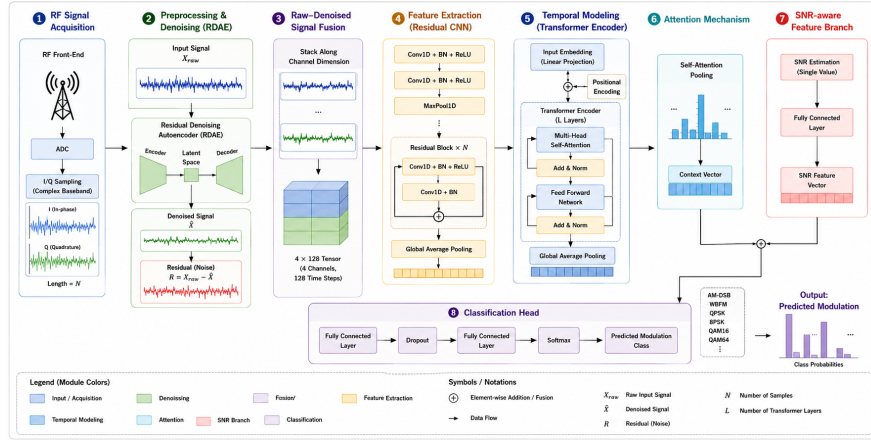


Fig. 1: Overall architecture of the introduced Adaptive RF Signal Recognition pipeline utilizing residual noise suppression, raw-denoised fusion, and Transformer encoder attention-driven learning.

3.3 Complete Signal Processing Workflow

The workflow begins with RF signal acquisition and I/Q sampling. The sampled signal is corrupted utilizing AWGN. The noisy RF signal is processed through the residual noise suppression module to generate denoised RF representations. The denoised representation is further fused with the original RF signal utilizing weighted residual fusion.

The denoised RF representation is combined with the original RF signal using weighted residual fusion to preserve both noise-robust and discriminative signal characteristics. The fused signal is computed as

$$X_{fused} = \alpha X_{denoised} + (1 - \alpha) X_{raw} \quad (5)$$

where X_{raw} denotes the original RF signal, $X_{denoised}$ represents the output of the residual denoising autoencoder, and α is the fusion coefficient. In our implementation, $\alpha = 0.40$, allowing the fused representation to retain complementary information from both the denoised and original signals while preserving important modulation characteristics.

Finally, the original RF signal and the weighted fused representation are concatenated along the channel dimension to construct the four-channel input tensor used by the proposed CNN-Transformer network. The weighted fused signal representation is then concatenated with the original raw signal to produce a four-channel fusion tensor representation: $X_{fusion} \in \mathbb{R}^{4 \times 128}$.

3.4 Residual CNN-Transformer encoder Recognition Backend

The backend is composed of multiple residual CNN blocks followed by Transformer encoder layers. Each residual CNN block contains 1D convolution with 64

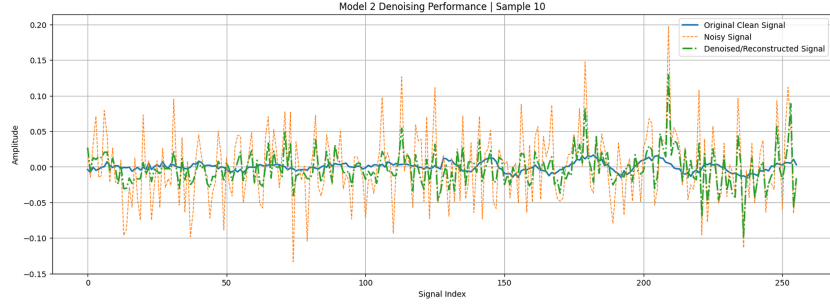


Fig. 2: Denoising performance visualization of the RDAE module comparing clean, noisy, and reconstructed RF signal samples.

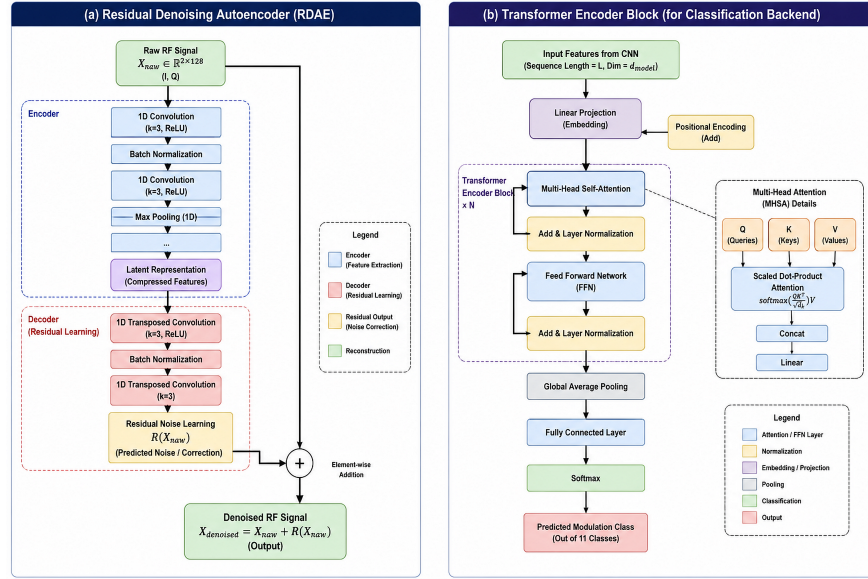


Fig. 3: Flowchart representation of the Residual Noise Suppression Autoencoder and Transformer encoder recognition backend.

to 128 filters, batch normalization, ReLU, and identity skip connections. Identity shortcuts are vital here to ensure that spatial features from the 4-channel signal representation flow directly to deeper layers.

Learned representations are then flattened and augmented with Positional Encodings to maintain sequence order before being processed utilizing multi-head self-attention. For a given signal representation sequence X , the Query (Q), Key (K), and Value (V) matrices are computed as:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \quad (6)$$

The attention-driven prediction result is then computed utilizing:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (7)$$

4 Evaluation Strategy

4.1 Benchmark Dataset

The RadioML 2016.10a benchmark dataset [1], [10] is utilized, containing 220,000 RF samples, 11 modulation classes, and SNR values from -20 dB to +18 dB. This study focuses on $\text{SNR} \geq -8$ dB.

4.2 Training Configuration

The proposed pipeline was realized using the PyTorch framework and trained in a Google Colab cloud environment utilizing an NVIDIA Tesla T4 GPU. To optimize training efficiency and reduce memory footprint, Automatic Mixed Precision (AMP) was employed via the GradScaler utility. Training was governed by the AdamW optimizer with a weight decay of 1×10^{-4} . A ReduceLROnPlateau scheduler was implemented to dynamically adjust the learning rate based on validation loss, ensuring stable convergence even under complex noise distributions.

Table 1: Hyperparameters

Parameter	Value
Optimizer	AdamW
Learning Rate	$5e-4$
Batch Size	256
Dropout (CNN/Transformer)	0.10 / 0.25
Weight Decay	$1e-4$
Epochs	50

5 Findings

The developed pipeline attains an overall recognition accuracy of 87.20%, surpassing baseline architectures. The reported overall recognition accuracy of 87.20% was computed over all test samples with SNR values ranging from -8 dB to +18 dB, thereby excluding extremely noisy conditions where reliable modulation discrimination is inherently difficult. The architecture illustrates high noise-resilience, particularly for cleaner RF condition schemes like QAM-64, where the Raw-Denoised fusion assists in the CNN filters identifying subtle amplitude

differences that are usually masked by Gaussian noise. The confusion matrix illustrates strong recognition predictive capability for most modulation classes, especially for higher-order modulation schemes like QAM16 and QAM64.

Table 2: Predictive capability Metrics

Metric	Value
Overall Recognition accuracy	87.20%
Macro Precision	0.89
Macro Recall	0.87
Macro F1-score	0.87
Weighted F1-score	0.87

Evaluation findings indicate that recognition predictive capability consistently gets superior as SNR levels increase. The pipeline keeps relatively stable predictive capability even in moderate noise situations, which illustrates the efficiency of the residual noise suppression and fusion strategy for reliable RF signal recognition.

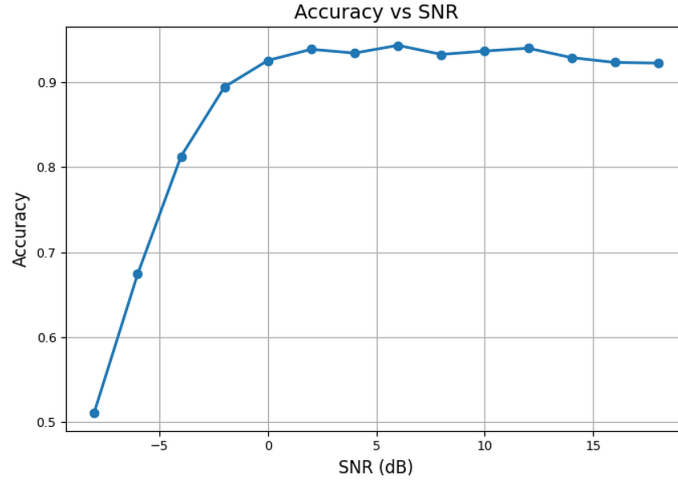


Fig. 4: Recognition accuracy of the developed pipeline as a function of Signal-to-Noise Ratio (SNR), illustrating consistent performance gains at higher SNR levels.

The Transformer encoder attention-driven visualization reveals that the self-attention mechanism efficiently captures significant long-range temporal relationships in RF signal sequences.

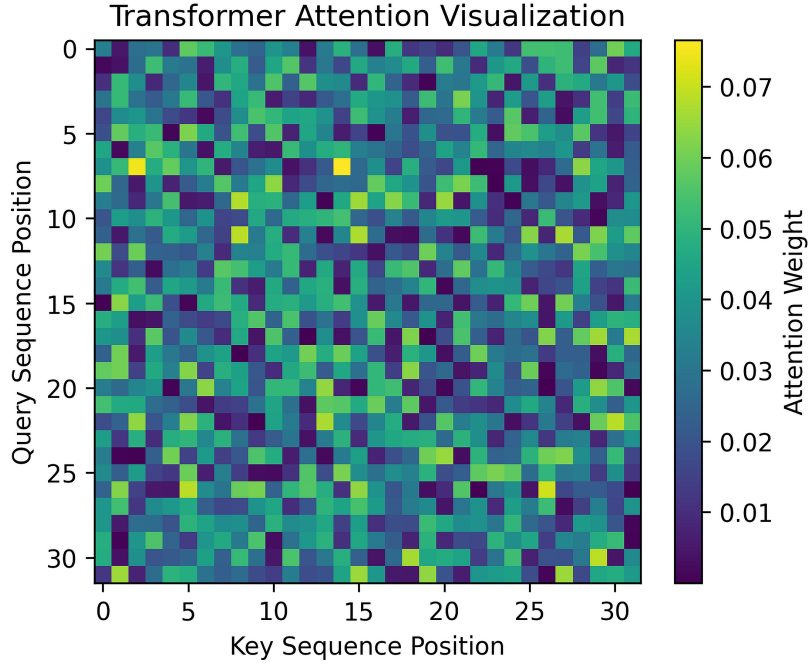


Fig. 5: Visualization of Transformer encoder attention-driven weights across RF signal sequences showing significant temporal dependencies.

5.1 Ablation Study

Table 3: Ablation Study Recognition accuracy

Architecture	Recognition accuracy
Baseline CNN	74.10%
CNN-BiLSTM-Attention	81.42%
CNN-BiLSTM-Attention (Fusion)	86.63%
Developed Residual CNN-Transformer encoder (Fusion)	87.20%

5.2 Limitations and Generalization

While the developed pipeline achieves strong results on the RadioML 2016.10a benchmark, certain limitations are acknowledged. The current evaluation is restricted to synthetic AWGN channels; the model’s robustness against complex

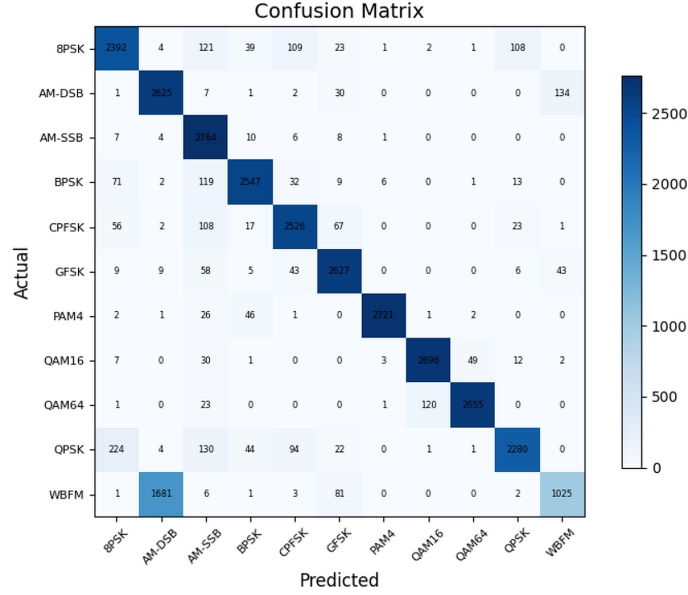


Fig. 6: Confusion matrix illustrating recognition predictive capability across all modulation classes.

Rayleigh fading and hardware-induced frequency offsets requires further over-the-air validation. Additionally, while the 2016.10a dataset serves as a robust baseline for initial validation, future work will involve scaling the architecture to larger-scale datasets like RadioML 2018.01a to ensure generalization across a wider array of modulation classes and higher-order schemes.

6 Computational Complexity Evaluation

The developed solution contains approximately 1.69 million trainable parameters. The average inference time for a single RF signal frame is 5.75 milliseconds when evaluated on an NVIDIA Tesla T4 GPU. This corresponds to a processing throughput of approximately 174 signal frames per second, demonstrating suitability for near-real-time automated modulation classification and providing potential for future deployment in Edge-AI and SDR-based environments.

7 Conclusion

Throughout this research, a study introduced an Adaptive RF Signal Recognition pipeline built upon Residual Noise Suppression and Raw-Denoised Signal Fusion utilizing a Residual CNN-Transformer encoder-Attention architecture. Evaluation findings on the RadioML 2016.10a benchmark dataset illustrate

that the developed pipeline noticeably enhances recognition predictive capability compared to conventional approaches, attaining 87.20% overall recognition accuracy along with strong macro-level performance assessment metrics. The developed pipeline illustrates strong potential for next-generation intelligent wireless communication architectures and real-time RF surveillance architectures. Future research will focus on the deployment of this fusion-learning pipeline onto Software Defined Radio (SDR) platforms for hardware-in-the-loop testing. Additionally, we aim to explore model compression and quantization techniques to enable the integration of this robust classifier into resource-constrained Edge-AI and IoT devices.

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